The Forced and the Chosen: The Tree of Reality's Declared Inputs, and the Plenitude of its Branches

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Abstract

The Tree of Reality generates the structures of physics from a minimal self-referential seed by one repeated operation. Most of its links are machine-checked; a handful are **declared** rather than forced. Its companion paper, *The Staked Gaps*, pre-registered the theory's **open problems**; this paper does the same for its **declared choices**, in three honest classes. **(1) Derivable postulates** — declared only because not-yet-derived — which we predict the mathematics will force (POSTULATE → GENUINE). **(2) Proven forks** — points where the generator is machine-checked to admit *coherent alternatives* (a different seed, signature, gauge cut, or depth) — which stay branches: this is the tree-of-life feature, not a flaw. **(3) Plenitude** — the prediction that **at every genuine fork, all coherent branches are realised** — *all that can be, becomes* — staked at its honest conjectural tier. We keep two distinctions sharp throughout, because blurring them would overclaim: a **convention** (chirality) is not a new world, and an open **completeness-question** (a possible fourth lineage) is not an alternative branch. The branches named here are predictions we have **not yet computed**; computing them — running the generator down each fork to see what physics it grows — is the sequel (*Tree v2*). Everything is dated and anchored to the Bitcoin blockchain.

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1. What this paper is

A generative theory earns its honesty from the sharpness with which it separates what it *forces* from what it *chooses*. *The Staked Gaps* pre-registered the first kind of incompleteness — the **open problems**: the two ◇ boxes and the partial links whose remainders are named, each predicted to close in a stated shape. This paper pre-registers the second kind — the **declared choices**: the steps the mathematics does *not* force, recorded in the capstone's audit as POSTULATE or carried as a chooser's constraint.

These are not gaps in the proofs; they are places where the generator, having grown a structure, does not single out one continuation. The thesis of this paper is that such places are not the theory's weakness but the exact sites of its most expansive claim: **where the tree genuinely forks, we predict every coherent branch is real.** A theory that forced every step would be a single chain; the Tree of Reality is a *tree* — by design, and (at four nodes) by machine-checked proof.

We state the choices in three classes, and we keep two honest distinctions sharp so the bold claim stays rigorous rather than loose.

2. Class 1 — Derivable postulates: predicted to be forced

A derivable postulate is declared *only because it has not yet been derived*; we predict the mathematics will one day force it, hardening its grade from POSTULATE to GENUINE. This class is **thin and we say so**: of the capstone's three POSTULATE links, only one belongs here honestly.

- **L14 — three generations.** The mathematics *is* proved: by Frobenius's theorem the real division algebras are $\mathbb{R}$, $\mathbb{C}$, $\mathbb{H}$; the pure (imaginary) quaternions form a three-dimensional space; the complexification identities hold at every size. What remains declared is the **identification** — that those three directions *are* the three fermion generations. We predict this identification earns a derivation. **Confidence: low, and honestly so** — it is an interpretive bridge, and the capstone proves the complexification cannot by itself decide the real form, so the postulate is genuinely load-bearing today. **Kill-condition:** a proof that the three-dimensional quaternionic structure does *not* determine the generation count (e.g. that the same construction equally permits a different number) refutes the prediction at this site.

Two POSTULATE links do **not** belong in this class, and placing them here would contradict the capstone's own results:

- **L7 — three lineages** is *not* a safe "predict-forced" candidate. The claim that End, Aut and the inner product are the *only* canonical lineages "has no formal statement; a fourth natural construction is already in use elsewhere in mathematics." Predicting "three, forced" could be *refuted*. We treat it in §5 as a completeness-question, not a forcing prediction.
- **L12 — chirality** is a **convention**: in the estate, the mirror statement compiles with the same one-line proof, so parity violation is *modelled, not derived*. A convention cannot be forced; we treat it in §5.

3. Class 2 — Proven forks: the branches

A proven fork is a node where the generator is **machine-checked to admit more than one coherent continuation** — genuinely distinct physics, not a relabelling. These are not predictions; they are theorems, and they are what makes the structure a tree. The campaign proved four:

- **The seed forks over the reals.** Over $\mathbb{R}$, two seeds qualify — $M_2(\mathbb{R})$ and the quaternions $\mathbb{H}$ — proved non-isomorphic, with the *same* complexification: two distinct, equally qualified starting points. *(`RealDivisionQuaternionCase`, `realForm_not_determined_two`.)*
- **The signature forks.** Three signatures share one Clifford algebra; the Riemannian worlds are not excluded. Lorentzian-over-Riemannian is a recorded choice, not a theorem. *(`LorentzianChosen.three_signatures_one_algebra`.)*
- **The size-sixteen level forks.** Every factorisation $a \cdot b \cdot c = 16$ decomposes M_{16}, and the mathematics prefers none; the Pati–Salam cut is our address at that ring, not its only tenant. *(link L11.)*
- **The depth is not forced.** The criterion that motivates reading physics at the fourth ring provably does not compel that depth. *(`CascadeMinimality.not_forall_eq_four`.)*

Each fork is a place the tree branches into other coherent mathematics. Which branch we inhabit is fixed not by derivation but by **matching observation** — ours is the branch with complex amplitudes, a Lorentzian clock, the Pati–Salam cut, read at the fourth ring, because that is the physics we measure.

4. Class 3 — Plenitude: at every fork, all branches are realised

Here is the paper's central prediction, and its boldest. For every *genuine fork* of §3, we predict:

> ***All coherent branches are realised. Wherever the generator admits more than one coherent continuation, every one of them is actual — in our universe as undiscovered sectors or regimes, or beyond it as other realisations. All that can be, becomes.***

This inherits, and states systematically, the programme's published Tier-3 conjecture (*The Infinite Ground*; the origin statement of *The Theory of Everything and the Origin of Reality*): *the space of actuality exhausts the space of possibility*. Its force here is that the forks are no longer an embarrassment of unforced choices — they are the **branch points of a real, exhaustively realised tree**. This makes the theory at once more *expansive* (reality is the

whole space of coherent branches, not one) and more *economical* (it dissolves the question "why this branch?": all branches exist; observation selects ours).

These are predictions we have not yet computed. At present the siblings are proved only *to exist* (the four theorems of §3). We predict that each, walked all the way up by the generator, grows a coherent alternative physics — and that all such coherent branches are realised. Computing them is the sequel (§7).

The honest tier. This is the most speculative class in the paper, and we label it so without hedging: plenitude is a **meta-empirical conjecture**, not a theorem, and not falsifiable in the ordinary way — the absence of a sibling from *our* observations refutes nothing, because that is exactly what the conjecture predicts. It is a bet, held loosely, placed on the record at the tier its source assigns it. What *is* falsifiable is narrower and stated in §6.

5. Two distinctions that keep the claim rigorous

Plenitude applies to **genuine forks** — distinct coherent physics. Two of the tree's declared inputs are *not* forks, and we refuse to smuggle them in, because doing so would make a false claim.

- **A convention is not a new world.** Chirality (L12) is a *relabelling*: the mirror theory is the *same physics* under a renamed handedness, proved by the identical one-line proof. There is no distinct "mirror universe" for plenitude to realise; there is one world with a conventional label. What may yet be forced is the *chiral structure itself* (that the theory is parity-violating at all); the *left/right name* is convention, permanently.
- **A completeness-question is not an alternative branch.** The possible fourth lineage (L7) asks whether *our own* branch has more canonical structure than the three we have mapped — not whether a different branch is realised elsewhere. If a fourth lineage exists, it is additional structure *here*, to be discovered, and we flag it as an open question with its own stake: **we predict the lineages are either shown exhaustive, or a fourth is found** — and either outcome is informative, neither is a plenitude branch.

6. Falsifiability, by class

Honesty requires stating, class by class, what could count against each prediction — and admitting where a class is a bet rather than a test.

- **Class 1 (L14):** falsifiable. A proof that the quaternionic structure does not determine the generation count refutes the forcing prediction.

- **Class 2 (the forks):** already theorems; not at issue. What *would* count is the sequel's risk: if a fork, walked up, is shown to grow *no* coherent physics (to collapse), then that branch is not a realisable world, and plenitude's domain shrinks accordingly.
- **Class 3 (plenitude):** a conjecture, not an ordinary test — a sibling's absence from observation refutes nothing. Its one honest hostage is §4's computational promise: were *every* non-trivial fork, once computed, to collapse into incoherence, plenitude would have nothing left to realise.

The boundary that keeps all of this principled is the one the capstone drew: **the tree is refutable on the branch it walks, and never excused by the branches it merely permits.** No failure on the walked branch may be reclassified, after the fact, as a sibling.

7. The sequel: computing the branches (Tree v2)

This paper *predicts* the branches are realised; it does not compute them. The natural next work — and the one that would turn plenitude from metaphysics toward mathematics — is to **run the generator down each fork and compute the physics it grows**: the quaternionic seed, the Riemannian signatures, the other size-sixteen cuts, the other depths. Each such branch is a full tree analysis in its own right, with its own cascade, its own gaps, and its own choices; the undertaking is a true second edition of the capstone, not an appendix. Its payoff is that it makes plenitude *live*: each fork, once walked, either grows a coherent alternative physics — strengthening the claim that the tree genuinely branches — or collapses, bounding it. We register the direction here and leave its execution to the rising capability that the companion paper's meta-prediction anticipates.

8. The meta-prediction

As with the gaps, we stake a forecast of closure, at its appropriate confidence: **as automated reasoning advances, the derivable postulate (L14) will earn its derivation; the completeness-question (L7) will be settled either way; and the sibling branches will be computed, one fork at a time.** Plenitude itself is not the kind of claim a proof closes — it is the reading those computations are placed within — and we keep it, permanently and visibly, at its conjectural tier.

9. Provenance and verification

This paper stakes predictions and states a conjecture; it proves nothing new. What is machine-checked is the framework it reasons from — the four fork-theorems of §3 and the postulate grades of §2 — audited link by link in the capstone and its public Lean estate (`github.com/`

`wonderben-code/convergence-codex`). These predictions sharpen, and do not contradict, the declared-choice accounting first registered in the capstone (28 September 2026); this document is dated, versioned on Zenodo, and anchored to the Bitcoin blockchain, so that the date of each prediction is independently checkable against the date of any future resolution.

Trust nothing; run everything.

References

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